THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 9 & 10: **Life Processes in Plants, Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Leaves are called the food factories of plants because they: 1

(a) produce seeds **(b)** carry out photosynthesis
(c) anchor the plant **(d)** absorb water

2. The raw materials for photosynthesis are: 1

(a) glucose and oxygen **(b)** carbon dioxide and water
(c) oxygen and water **(d)** nitrogen and water

3. Before testing a leaf for starch, chlorophyll is removed using: 1

(a) cold water **(b)** lime water **(c)** salt solution **(d)** warm alcohol

4. Water enters the plant roots through: 1

(a) flowers **(b)** stem **(c)** root hairs **(d)** leaves

5. Respiration in plants releases: 1

(a) minerals **(b)** starch **(c)** energy **(d)** chlorophyll

6. The Moon is visible because it: 1

(a) absorbs light **(b)** reflects sunlight
(c) produces its own light **(d)** is transparent

7. Which is opaque? 1

(a) clear water **(b)** clear glass **(c)** air **(d)** wooden board

8. As an object moves closer to the light source, its shadow becomes: 1

(a) invisible **(b)** same **(c)** larger **(d)** smaller

9. In a plane mirror, the image is of ___ size as the object. 1

(a) smaller **(b)** half **(c)** same **(d)** double

10. A pinhole camera forms an image that is: 1

(a) coloured only **(b)** erect
(c) inverted **(d)** bigger than object always

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. What are parasitic plants? Give an example. 2

12. How do insectivorous plants get nitrogen? 2

13. What is a pinhole camera? 2

14. Why is the pinhole image inverted? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why is Cuscuta called a parasite? 3

16. Why is the Sun's light able to reach Earth although there is no air in space? 3

17. Why do we say plants are producers? 3

OR

Describe the experiment on the role of water in plant growth. 3

18. List the activities to classify materials as transparent, translucent or opaque. 3

OR

Why does a dark room seem bright when a torch is switched on and a mirror used? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Why are legumes called nitrogen fixers? 5
(ii) Role of Rhizobium.
(iii) Importance to farmers.

OR

Case study: A pitcher plant grows in nitrogen-poor soil. 5
(i) Why does it trap insects?
(ii) How?
(iii) Does it photosynthesise?
(iv) Classify.

20. (i) Differentiate shadow and image. 5
(ii) Differentiate plane-mirror image and pinhole image.
(iii) Which is real?

OR

Case study: Light from the Sun forms a round spot below a tree through gaps between leaves. 5
(i) Why round?
(ii) Which device works similarly?
(iii) Would size change with gap?
(iv) What does this show about light?

END OF QUESTION PAPER — Total: 40 Marks

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